

IoT-Based Smart Driver Drowsiness Detection and Vehicle Safety System Using Raspberry Pi

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Abstract—Driver drowsiness is a major factor contributing to road accidents worldwide. Reduced alertness, slower reaction time, and temporary loss of concentration significantly increase the probability of collisions.

This paper presents an IoT-based smart driver drowsiness detection and vehicle safety system using Raspberry Pi, OpenCV, Arduino Uno, and a camera module.

The proposed system continuously monitors the driver's eye state and yawning behavior using computer vision techniques and detects fatigue based on eye closure duration and facial analysis.

When drowsiness is detected, alerts are generated using a buzzer, hazard LED indicator, Telegram notification system, and motor speed reduction mechanism.

The proposed solution is low-cost, real-time, and suitable for embedded safety applications.

The project aligns with Sustainable Development Goal (SDG) 3: Good Health and Well-Being by helping reduce fatigue-related road accidents.

Index Terms—IoT, Driver Drowsiness Detection, Raspberry Pi, OpenCV, Arduino Uno, Embedded Systems, Road Safety, Computer Vision

I. INTRODUCTION

Road accidents are a serious global safety issue, causing nearly 1.19 million deaths every year according to the World Health Organization (WHO).

A significant percentage of these accidents occur due to driver fatigue and reduced alertness. Drowsiness affects the driver's ability to react quickly, reduces concentration, and may cause microsleep events that lead to severe accidents.

Recent developments in Internet of Things (IoT) technology and embedded systems have enabled real-time monitoring of driver behavior using low-cost hardware.

Computer vision techniques allow continuous observation of facial features such as eye closure, blinking frequency, and yawning patterns.

The specific problem addressed in this work is the real-time detection of driver drowsiness using a low-cost IoT-based embedded system without requiring wearable sensors.

The proposed system uses Raspberry Pi with OpenCV algorithms to process live video frames. Image frames are analyzed for eye closure and yawning patterns.

When fatigue is detected, alerts are generated using a buzzer and LED indicator. Telegram cloud notifications are also transmitted using Wi-Fi communication.

Additionally, the system demonstrates intelligent vehicle safety control through Arduino-controlled motor speed reduction.

The project aligns with Sustainable Development Goal (SDG) 3: Good Health and Well-Being by helping reduce accidents caused by driver fatigue.

II. RELATED WORK

Several studies have proposed driver drowsiness detection systems using image processing and deep learning techniques.

Kim et al. developed a real-time driver monitoring system using facial landmark detection and infrared cameras.

Lamaazi et al. proposed an edge-based fatigue detection system using convolutional neural networks.

Phan et al. presented an IoT-based fatigue detection framework using deep learning models.

Most existing systems require expensive hardware or computationally intensive algorithms.

Therefore, there is a need for a low-cost embedded system capable of real-time operation.

III. SYSTEM DESIGN AND ARCHITECTURE

A. System Overview

The proposed system is an IoT-based driver drowsiness detection and vehicle safety system developed using Raspberry Pi, OpenCV, Arduino Uno, and alert modules.

The system continuously monitors the driver's facial features using a USB camera connected to the Raspberry Pi.

Computer vision algorithms process the live video stream to detect eye closure duration and yawning patterns.

Based on the calculated drowsiness level, alerts are generated through a buzzer, LED indicator, Telegram notifications, and simulated vehicle motor control.

B. Block Diagram

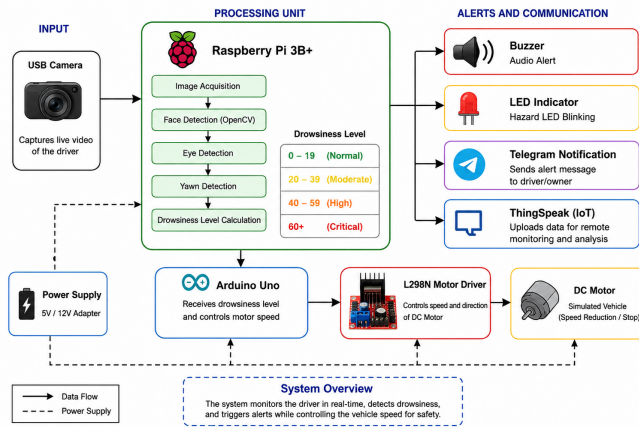


Fig. 1. System Block Diagram

C. Circuit Diagram

The circuit design of the proposed system consists of Raspberry Pi 3B+, Arduino Uno, L298N motor driver, USB camera, buzzer, LED indicator, and DC motor.

The Raspberry Pi acts as the main processing unit and performs real-time image processing using OpenCV algorithms.

A USB camera connected to the Raspberry Pi captures continuous video frames of the driver.

The Arduino Uno is connected to the Raspberry Pi through USB serial communication.

Based on the calculated drowsiness level, commands are transmitted from the Raspberry Pi to the Arduino.

The Arduino controls the L298N motor driver using PWM signals.

The motor driver regulates the speed of the DC motor according to the detected drowsiness level.

During critical drowsiness conditions, the motor speed gradually decreases and eventually stops.

A buzzer is connected to the Raspberry Pi GPIO pins to provide immediate audio alerts.

An LED indicator connected to the Arduino acts as a hazard warning light and blinks during dangerous drowsiness conditions.

The complete circuit diagram of the proposed system is shown in Fig. 2.

D. Hardware Components

E. Software Architecture

The software architecture consists of:

- Image Acquisition Layer
- Image Processing Layer
- Alert Generation Layer
- Vehicle Safety Layer

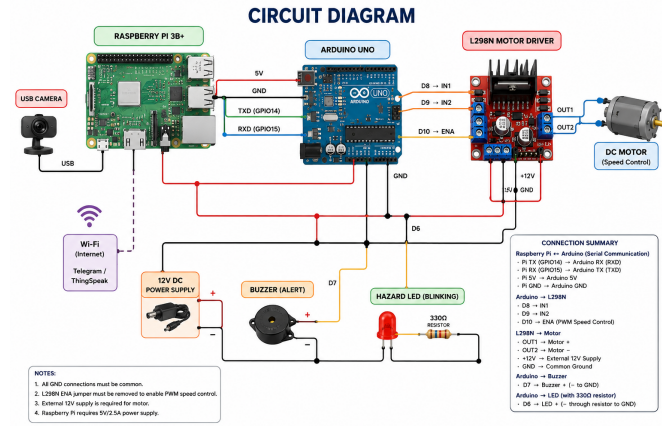


Fig. 2. Circuit Diagram of the Proposed System

TABLE I
HARDWARE COMPONENTS

Component	Purpose
Raspberry Pi 3B+	Main processing unit
USB Camera	Captures driver video
Arduino Uno	Motor control
L298N Driver	Motor speed control
DC Motor	Vehicle simulation
Buzzer	Audio alert
LED	Hazard indication

IV. METHODOLOGY

A. Working Methodology

The system continuously monitors the driver's facial activity using a USB camera connected to the Raspberry Pi.

Each video frame is processed using OpenCV image processing techniques.

The methodology includes:

- 1) Capture live video frame
- 2) Convert frame into grayscale
- 3) Detect face
- 4) Detect eyes
- 5) Detect yawning
- 6) Calculate drowsiness score
- 7) Trigger alerts
- 8) Send Telegram notification
- 9) Reduce motor speed

B. System Flow

- Continuous real-time monitoring using a USB camera connected to the Raspberry Pi
- Capture live video frames from the driver
- Convert captured frames into grayscale images for faster processing
- Detect face region using Haar Cascade classifiers
- Detect eye regions for fatigue analysis

- Monitor eye closure duration continuously
- Detect yawning behavior using facial analysis
- Dynamically calculate drowsiness score
- Compare drowsiness score with predefined threshold values
- Generate buzzer alerts during drowsiness detection
- Activate hazard LED blinking mechanism
- Send Telegram notification for remote monitoring
- Reduce DC motor speed according to drowsiness level
- Completely stop motor during critical drowsiness conditions

C. System Flow Diagram

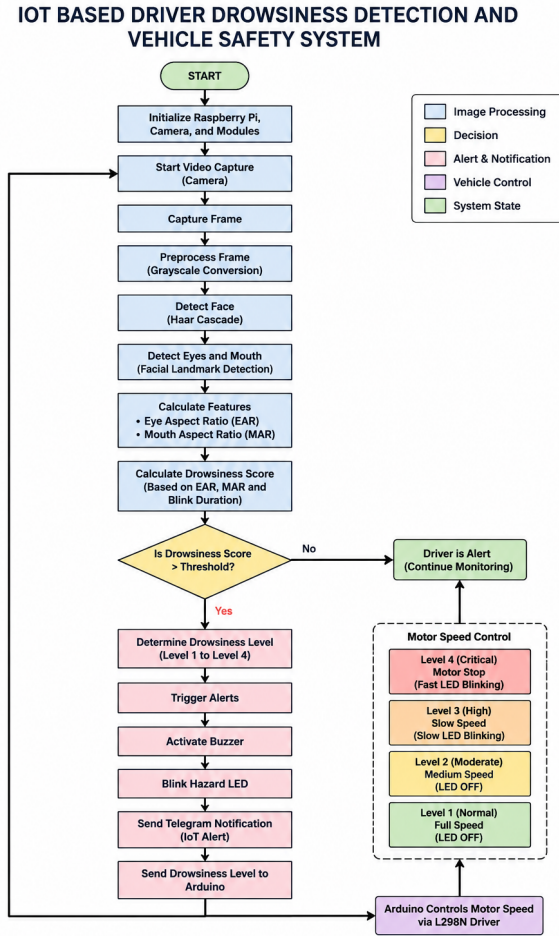


Fig. 3. System Flow Diagram of the Proposed Driver Drowsiness Detection System

D. Hardware Prototype

E. Output Screenshots

F. Alert Threshold Levels

V. RESULTS AND DISCUSSION

A. Experimental Results

The proposed system successfully performed:

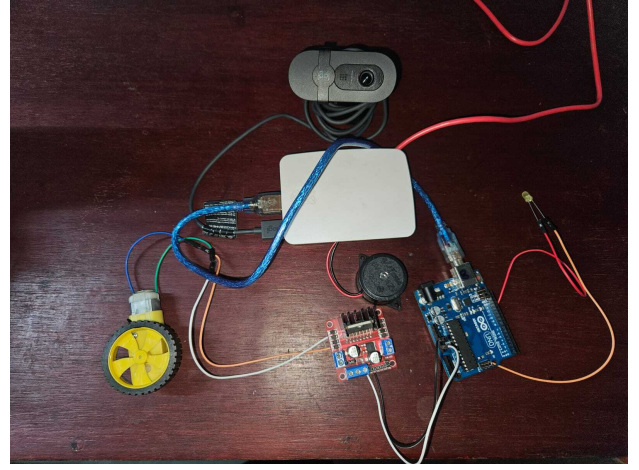


Fig. 4. Hardware Prototype

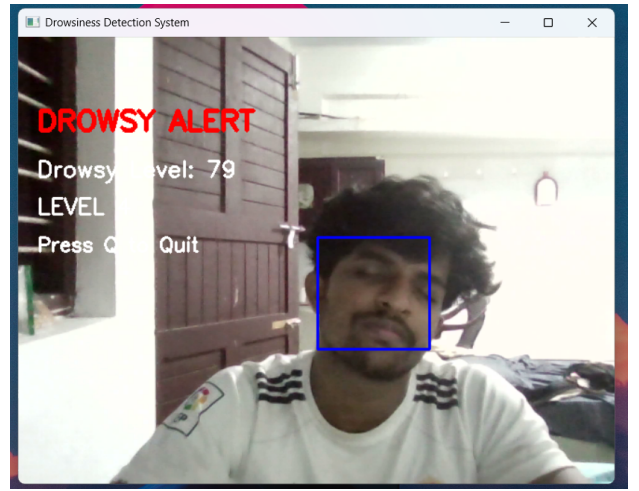


Fig. 5. Live Drowsiness Detection Output

- Real-time face detection
- Eye closure detection
- Yawning detection
- Drowsiness level calculation
- Telegram notification
- Buzzer alert
- Motor speed reduction
- Hazard LED blinking

B. Telegram Alert System

The developed system integrates Telegram-based IoT communication for remote driver monitoring.

When the detected drowsiness level exceeds the predefined threshold, the Raspberry Pi automatically sends an alert notification to the registered Telegram account using the Telegram Bot API.

The alert message includes warning information regarding the driver's fatigue condition and detected drowsiness level.

This feature enables real-time remote monitoring and improves emergency response capability.

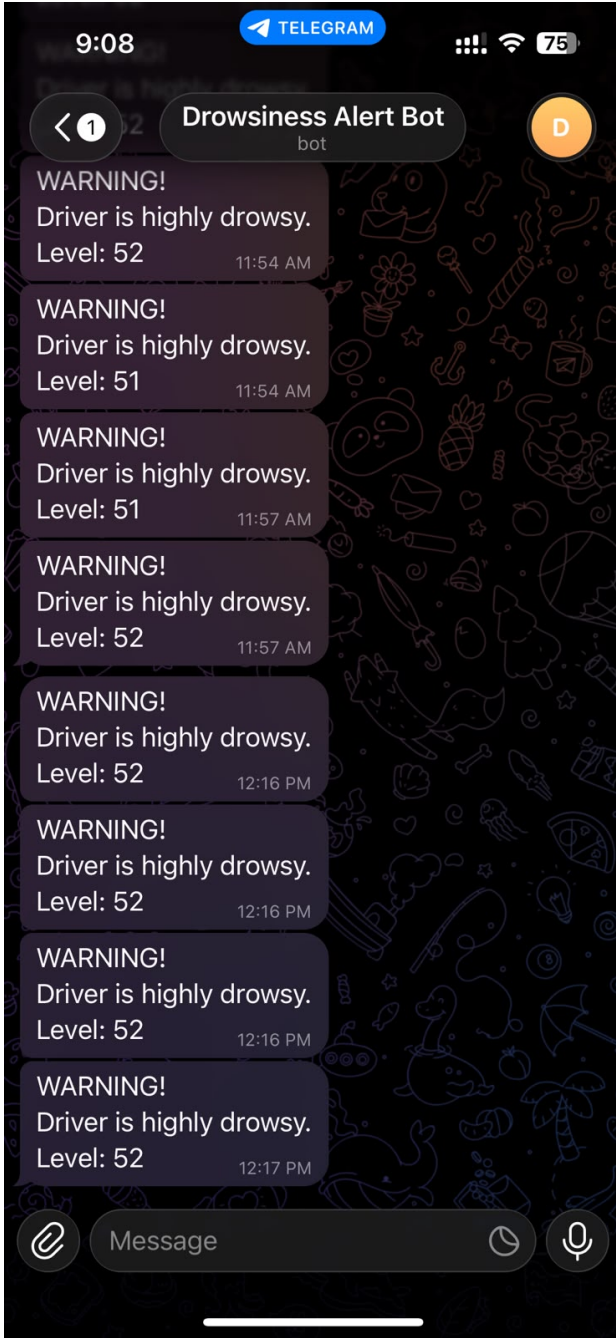


Fig. 6. Telegram Alert Notification

Fig. 6 shows the Telegram alert notification generated by the system.

C. Motor Control Results

D. Performance Analysis

E. Comparison with Existing Systems

F. Discussion

The proposed system demonstrates that low-cost embedded platforms such as Raspberry Pi can effectively perform real-time drowsiness detection using computer vision techniques.

TABLE II
ALERT THRESHOLD LEVELS

Drowsiness Level	System Action
0–19	Normal operation
20–39	Moderate warning
40–59	High drowsiness
60+	Critical alert and motor stop

TABLE III
MOTOR AND LED BEHAVIOR

Level	Motor State	LED State
LEVEL 1	Full Speed	OFF
LEVEL 2	Medium Speed	OFF
LEVEL 3	Slow Speed	Slow Blink
LEVEL 4	Motor Stop	Fast Blink

The integration of buzzer alerts, LED indicators, IoT notifications, and vehicle speed reduction improves overall driver safety.

G. Limitations

The system has certain limitations:

- Reduced accuracy under poor lighting
- Difficulty detecting eyes with spectacles
- Face angle sensitivity
- Dependence on camera quality

H. Future Improvements

Future enhancements may include:

- Deep learning-based facial landmark detection
- Infrared night vision cameras
- Mobile application integration
- GPS emergency alerts
- Real vehicle deployment
- Cloud analytics dashboard

VI. CONCLUSION

This work presented an IoT-based smart driver drowsiness detection and vehicle safety system using Raspberry Pi, OpenCV, Arduino Uno, and IoT communication.

The system successfully detects drowsiness by monitoring eye closure duration and yawning behavior in real time.

The developed prototype generates immediate alerts using a buzzer and LED indicator while also sending Telegram notifications for remote monitoring.

The integration of Arduino-based motor speed control demonstrates an intelligent vehicle safety mechanism capable of reducing vehicle movement during dangerous fatigue conditions.

The proposed system offers a low-cost, practical, and easily deployable solution suitable for smart transportation and driver safety applications.

TABLE IV
PERFORMANCE METRICS

Parameter	Result
Face Detection Accuracy	90%
Eye Detection Accuracy	85%
Alert Response Time	≤ 2 seconds
Telegram Delay	2–4 seconds
Processing Speed	Real-time

TABLE V
COMPARISON WITH EXISTING SYSTEMS

Feature	Existing Systems	Proposed System
Cost	High	Low
Complexity	High	Moderate
Deep Learning	Required	Not Required
Cloud Alert	Limited	Telegram + IoT
Vehicle Control	Limited	Available

The project aligns with Sustainable Development Goal (SDG) 3: Good Health and Well-Being by helping reduce fatigue-related road accidents and improving transportation safety.

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